

d) The column space of a matrix A is the set of solutions of $Ax=b$.

Solution: By Defn, Column space of a matrix A represents the set of all linear combinations of the columns of A . By Defn. of Ax , this represents the set of all possible Ax . $\therefore$ The given statement is False.

e) If B is an echelon form of a matrix A , then the pivot columns of B form a basis for $\text{Col } A$.

Solution: From the Warning of page 152, the given statement is False. $\square$

Exercises 23-26 display a matrix A and an echelon form of A . Find a basis for $\text{Col } A$ and a basis for $\text{Nul } A$.

$$23. A = \begin{bmatrix} 4 & 5 & 9 & -2 \\ 6 & 5 & 1 & 12 \\ 3 & 4 & 8 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 6 & -5 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution: The entries in Row 1 Column 1 and Row 2 Column 2 are pivots. $\therefore$ Columns 1 and 2 are pivot columns. By Theorem 13, the columns $\begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix}$ are basis for $\text{Col } A$.

$$\begin{bmatrix} 1 & 2 & 6 & -5 & 0 \\ 0 & 1 & 5 & -6 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{-2xRow2}]{\text{Row1=Row1}} \begin{bmatrix} 1 & 0 & -4 & 7 & 0 \\ 0 & 1 & 5 & -6 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The corresponding system of equations are: $x_1 - 4x_3 + 7x_4 = 0$
 $x_2 + 5x_3 - 6x_4 = 0$
 $0 = 0$.

$$\begin{aligned} x_1 &= 4x_3 - 7x_4 \\ x_2 &= -5x_3 + 6x_4 \end{aligned} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -7 \\ 6 \\ 0 \\ 1 \end{bmatrix}$$

Since x_3 and x_4 are free, x can be represented as the set of all linear combinations of the vectors $\begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -7 \\ 6 \\ 0 \\ 1 \end{bmatrix}$. Also it's easy to verify that these two vectors are linearly independent.